

TEST DATASET VALIDATION REPORT

AUTOMATED DETECTION OF GLACIAL LAKES FROM MULTI-TEMPORAL SATELLITE IMAGERY (GLOFEAGLES)

Submitted by

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1. Validation Dataset Details:

The given test dataset for validation of model contained images which were split into 6 class types. Each class represents a hindrance and a total of 2,226 images were given along with ground truth segmentation mask labels.

Class Type	Number of Images
Cloud Cover	12
Debris Cover	234
Moraine Dammed	841
Snow Cover	834
Terrain Shadow	239
Varying Turbidity	66
Total	2,226

2.Evaluation Result Details:

Evaluation Metric Class type	Accuracy	Mean IoU	Cohen' s Kappa	Precision	Recall	F1/Dice
Debris Cover	0.9945	0.6687	0.7580	0.8141	0.7592	0.7602
Moraine Dammed	0.9909	0.6688	0.7456	0.8080	0.7570	0.7491
Terrain Shadow	0.9958	0.6687	0.7466	0.8376	0.7370	0.7481
Varying Turbidity	0.9955	0.6974	0.7886	0.8597	0.7774	0.7907
Snow Cover	0.9939	0.6478	0.7315	0.8004	0.7356	0.7335
Cloud Cover	0.9980	0.7817	0.8684	0.9487	0.8168	0.8694
Average Metric values	0.9948	0.6888	0.7731	0.8448	0.7638	0.7752

Inference on Evaluation:

- **Pixel Accuracy (99.48%)** indicates near perfect global background matching but this occurs due to the fact that background(non-lake) area is massive compared to lake areas.

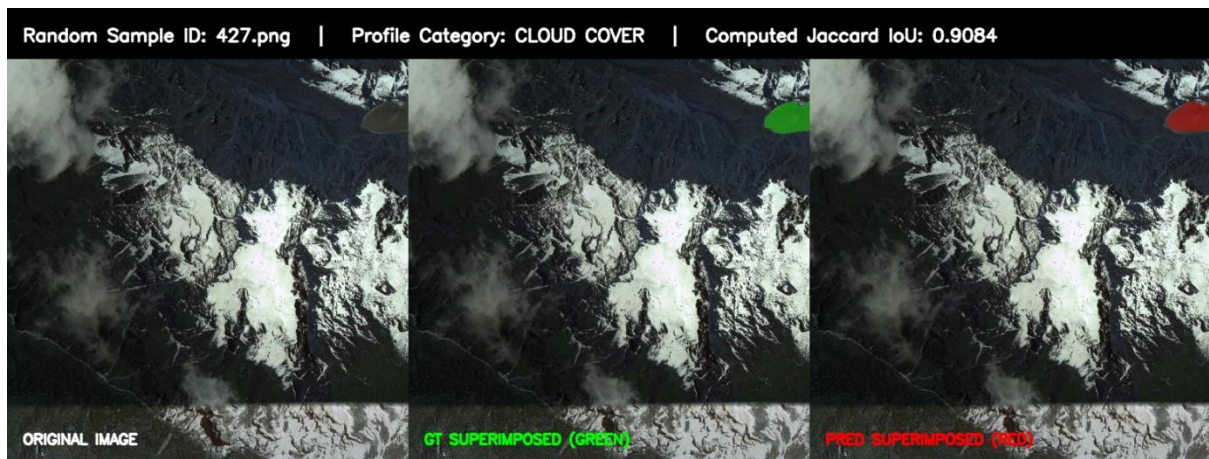
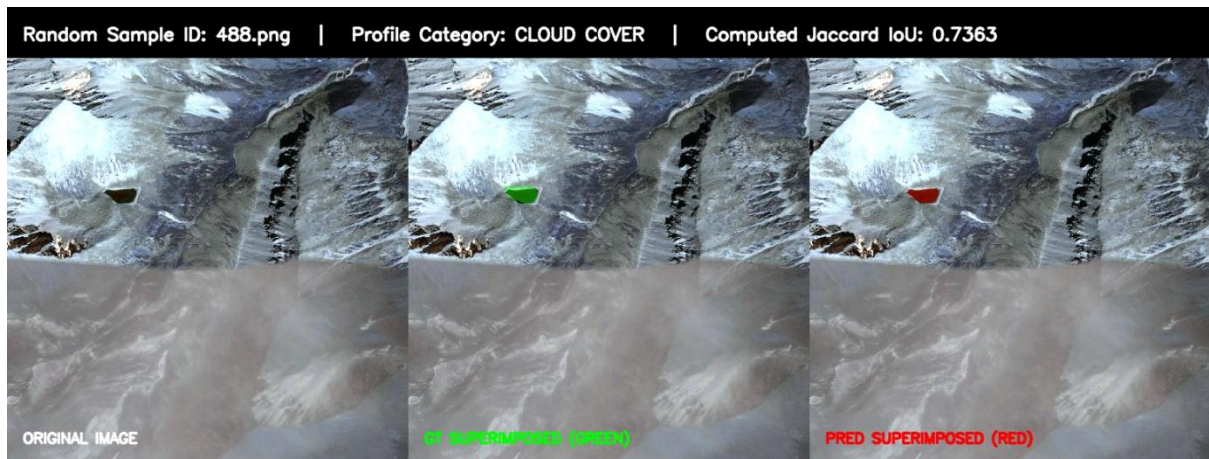
- **Precision (84.48)** indicates that the model is highly trustworthy, but misclassifying tricky mountain shadows (which appear as sometime occur with same texture as water) or dark terrain as water.
- **Recall (76.38%)** highlights the main bottleneck where the network plays it safe, occasionally missing tiny speck of ponds or shallow lake borders.
- **F1-Score / Dice (77.52%)** highlights a robust, high-performing balance between avoiding false positives and capturing actual target lakes.
- **Mean IoU (68.88%)** tells that the exact geometric perimeters and shapes drawn by the model align well with real physical boundaries and tiny specks of missing glacier lakes pulls the mean IoU. As this is strict pixel-to-pixel match between prediction and ground truth, the value is lesser compared with other metrics.
- **Cohen's Kappa (0.7731)** implies "substantial agreement" and confirming the model relies on true spatial feature learning rather than lucky guessing.

3.Sample Output of Model predictions:

The below shows images where the model performs well on critical obstacles and efficient than human estimation.

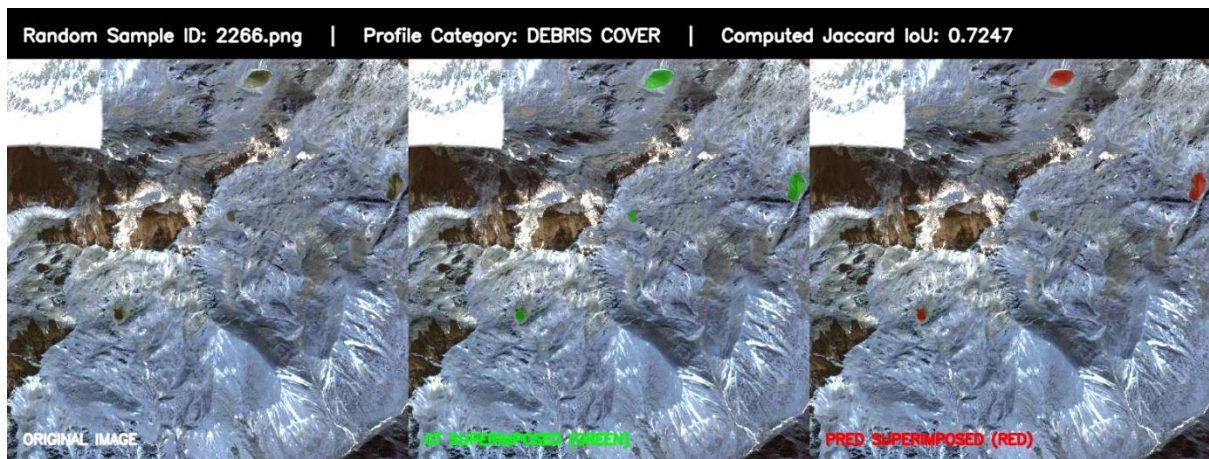
i)Cloud Cover:



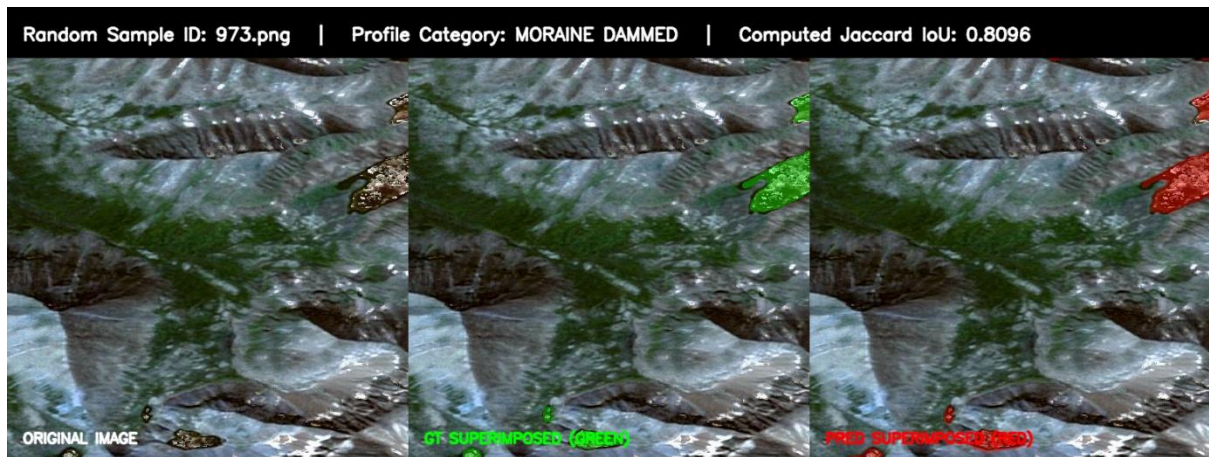


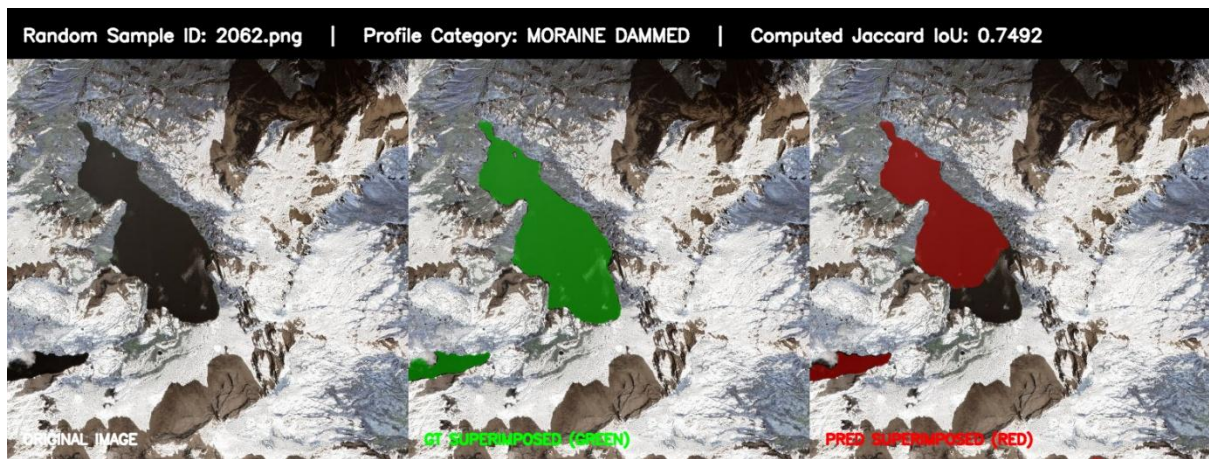
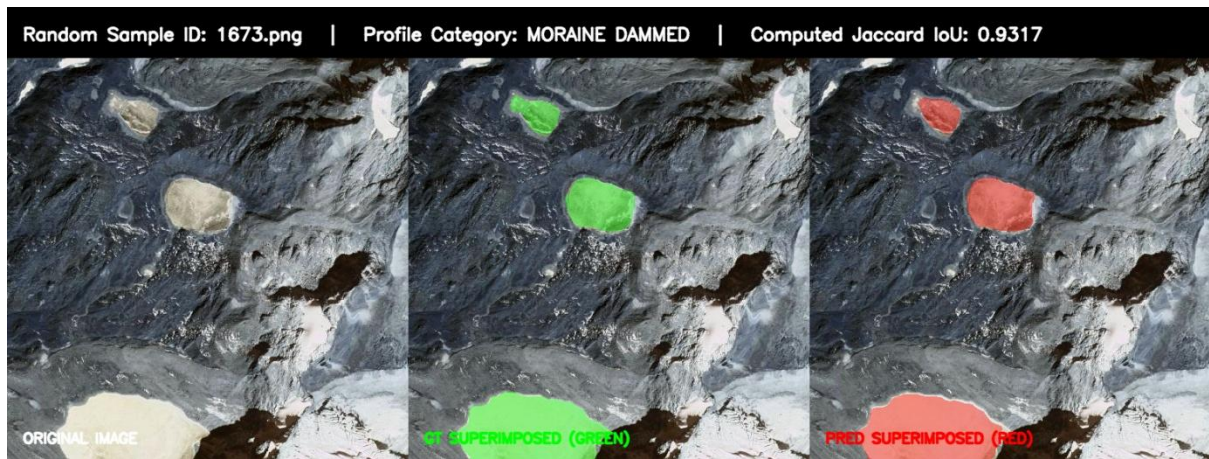
ii) Debris Cover:



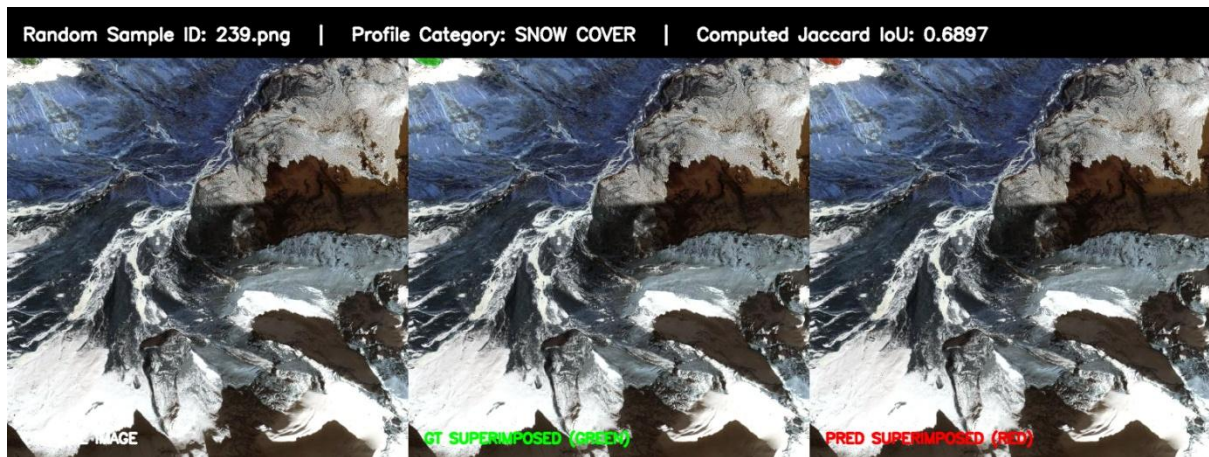


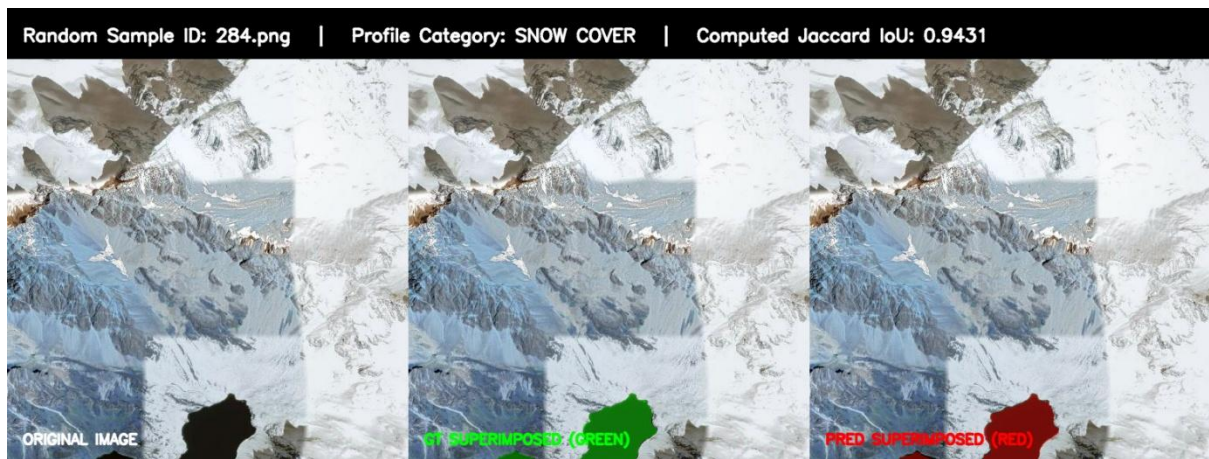
iii) Moraine Dammed:





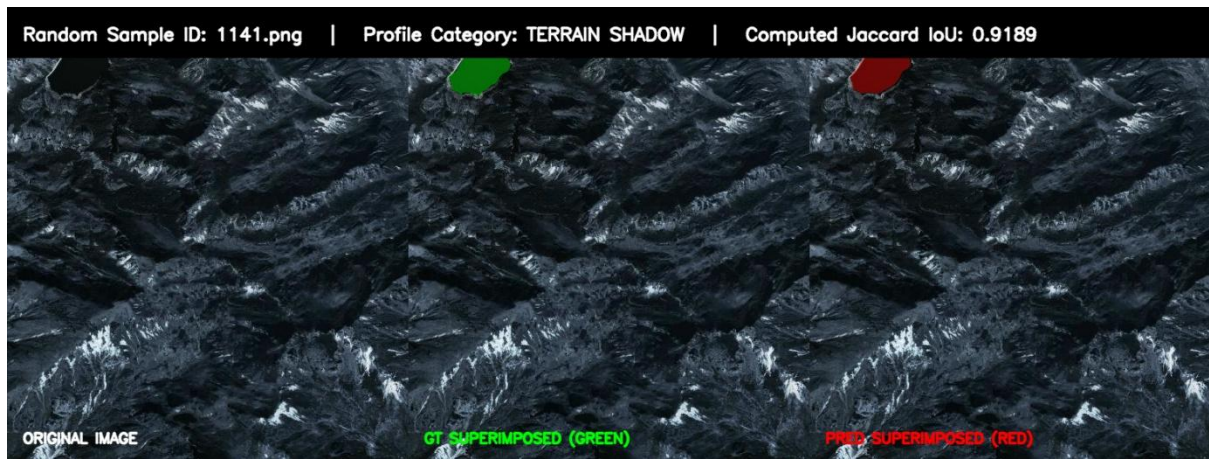
iv) Snow Cover:



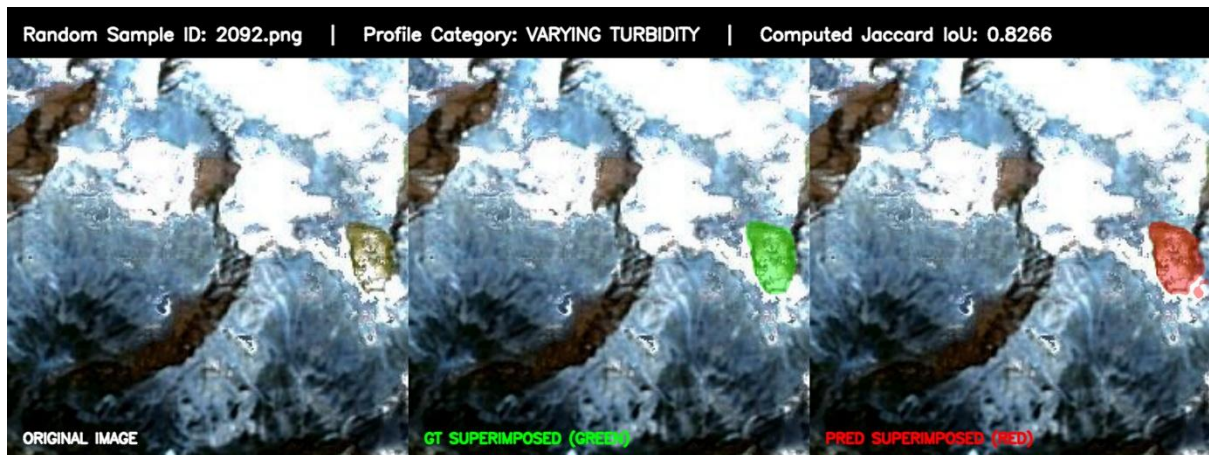


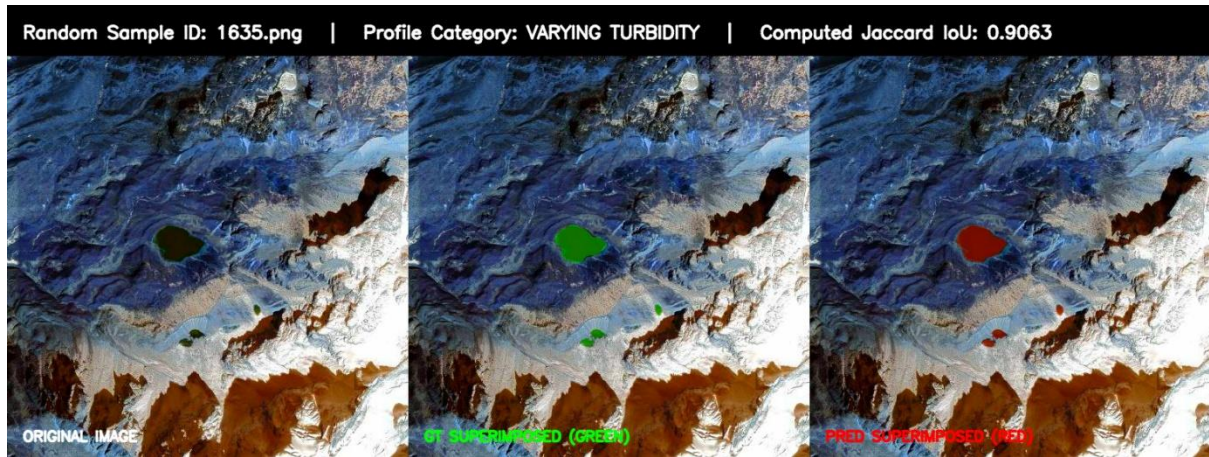
v) Terrain Shadow:





vi) Varying Turbidity:

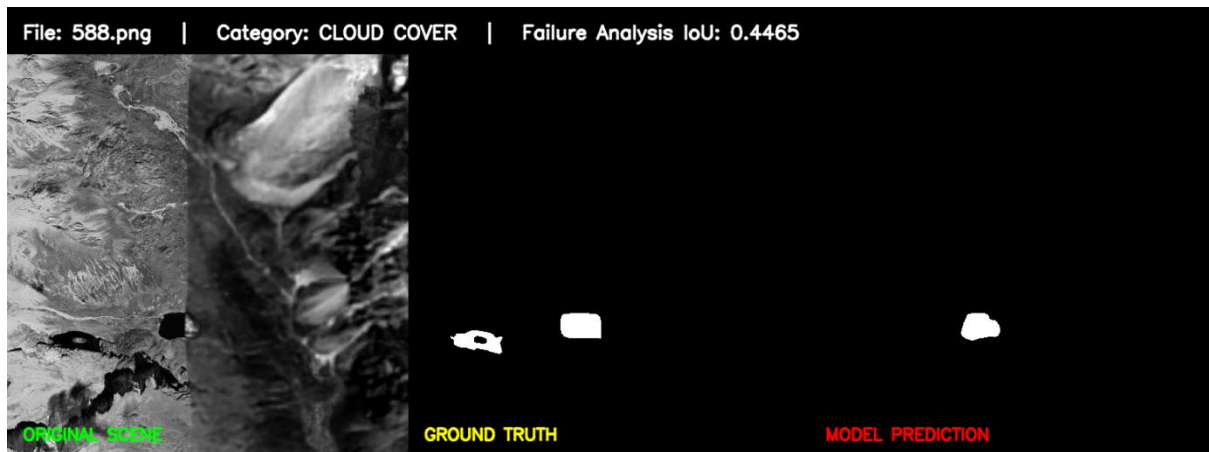




4.Failure Cases and Analysis:

Top 10 worst case images from each class type have been assessed and presented as follows having 0.00 IoU with opt reasons, which helps to know where the model struggles to identify glacier lakes.

i)Cloud Cover:



Possible reasons: Confusion between the cloud shadow and glacier lake and additional incorrect coverage of ground truth.

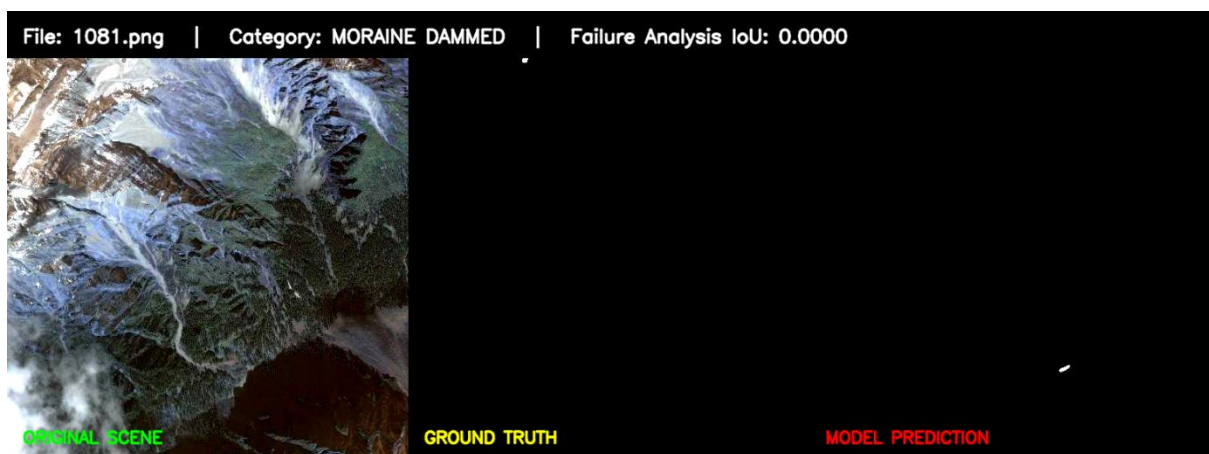
ii)Debris Cover:





Possible reasons: Lake regions which appears as partial at the edge of image over exposure image which makes the model to get confuses with background.

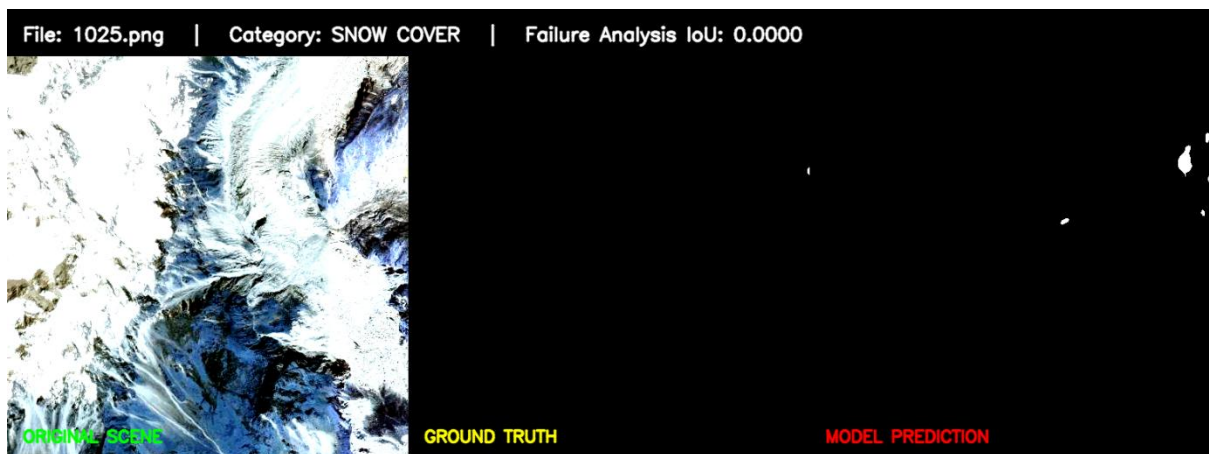
iii) Moraine Dammed:





Possible reasons: Tiny specks of glacier lake which in above images appears having incorrect ground truth labels with respect to actual images. High exposure images such as 1122.png hides the lake.

iv) Snow Cover:



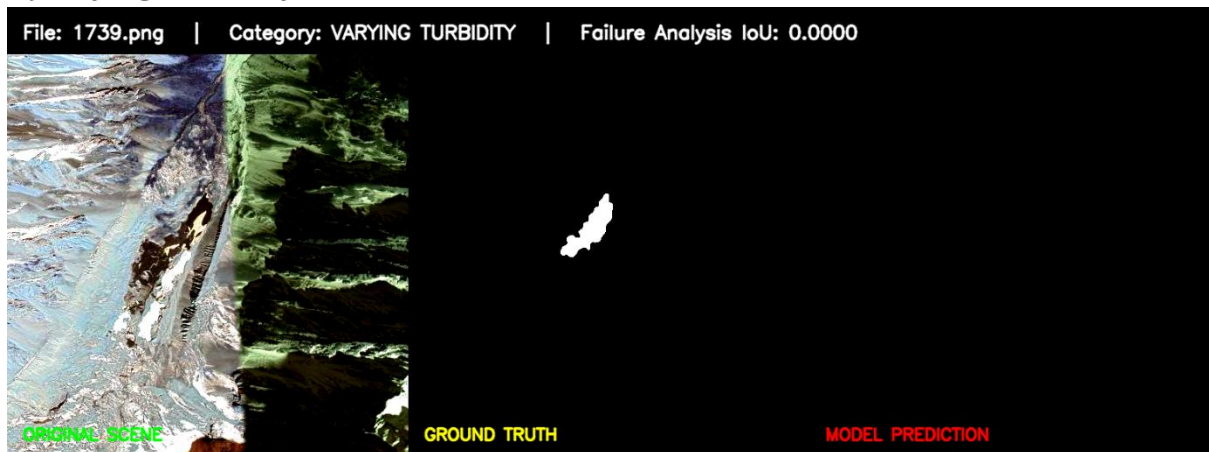
Possible reasons: Tiny specks of glacier lake with respect to actual images.

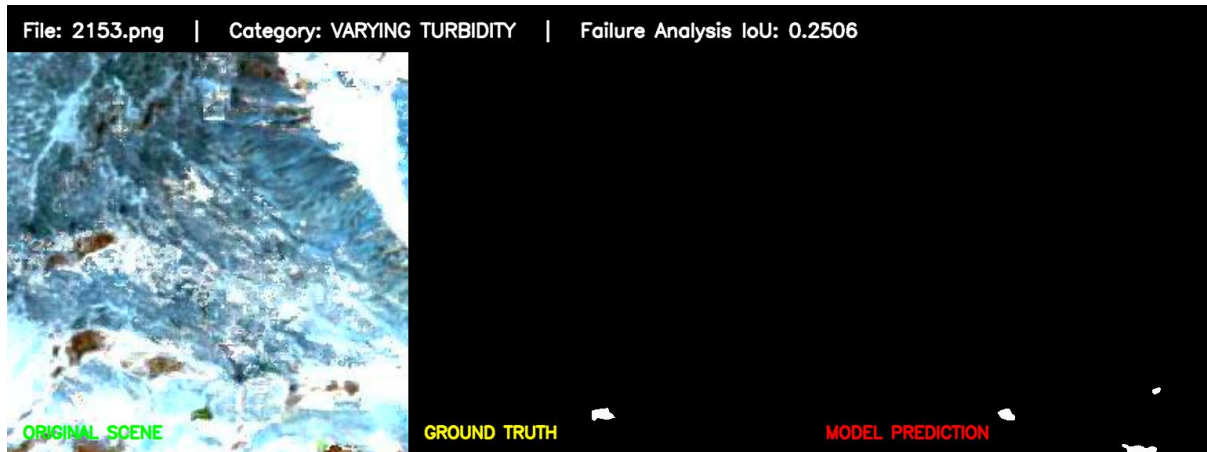
v)Terrain Shadow



Possible reasons: Tiny specks of glacier lake with respect to actual images and confusion with deep pitch-black shadows.

vi)Varying Turbidity





Possible reasons: Glacier Lake which have different texture due to presence of multiple colors and components, which is not common in other class types.